

Wearable Technologies and Applications (Wearable Informatics)

Winfree

Lecture 5

Let's Compare

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Unsupervised and
Supervised
Statistical Learning

Let's Compare

Real World Data

Linear Regression

Clusters

Classification

K-Means

K-Nearest-Neighbors

Comparing K-Means and
KNN (Similarities,
Differences?)

- ▶ Unsupervised - we do not provide labels, or human defined information about the “groups” or “status” of the data.
- ▶ Supervised - we label the data, giving exemplars of what those data represent. This human input makes use of *a posteriori* knowledge and / or *a priori* assumptions.

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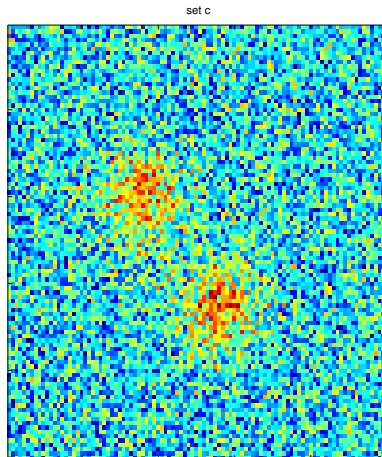
K-Nearest-Neighbors

Comparing K-Means and
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- ▶ *a priori* - knowledge, justifications, or arguments that exist independently from experience. Examples include mathematics ($3 + 2 = 5$), tautologies (all bachelors are unmarried), and deduction from pure reason (ontological proofs).
- ▶ *a posteriori* - knowledge, justifications, or arguments that depend on experience or empirical evidence. Examples include most aspects of science and personal knowledge.

Real World Data

Suppose we are looking at a dataset like this.
But we never *really* get the whole data set, representing everything...



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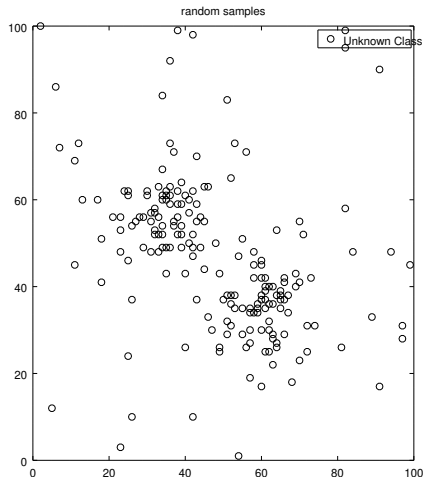
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Real World Data

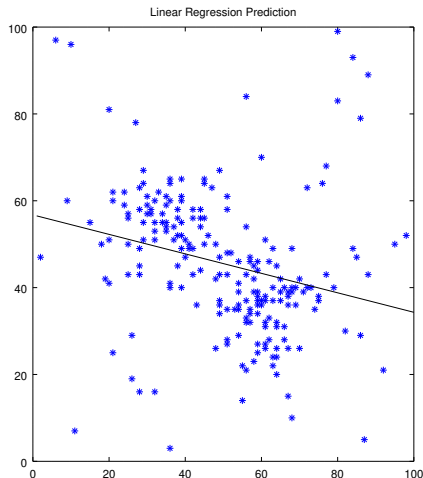
But we probably are actually only getting something like this.
How would we tackle this data set?
Assume that we don't have known classes or groups.



Linear Regression

We could regress it, with linear regression.

This tries to relate the y to the x . Depending on x , we will try to predict the y values.



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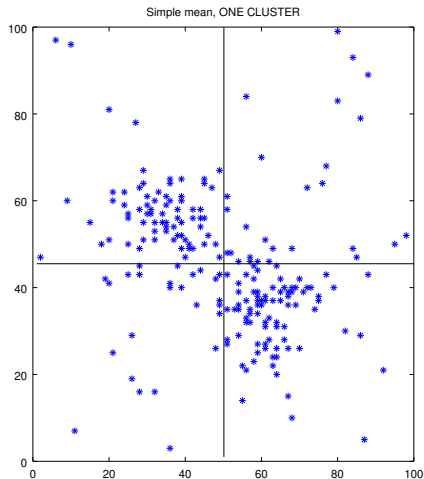
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Clusters

Or if we step back, we can see that this kind of looks like a cluster of data.
Call it a cluster?
But really it looks like there are two clusters there.



K-Means

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- ▶ k-means is an unsupervised learning method
- ▶ nothing is known about classification of data points in the group
- ▶ we seek to identify groups that are similar
- ▶ what set of points has an affinity to others?

K-Means How-To

1. randomly sample N data points from the source set
2. assign each of those to a cluster group
3. while (data remains)
 - 3.1 randomly select one point
 - 3.2 measure the distance to the centroids of the clusters
(mean of x and y cluster points)
 - 3.3 assign to the closest cluster
 - 3.4 remove that data point and repeat

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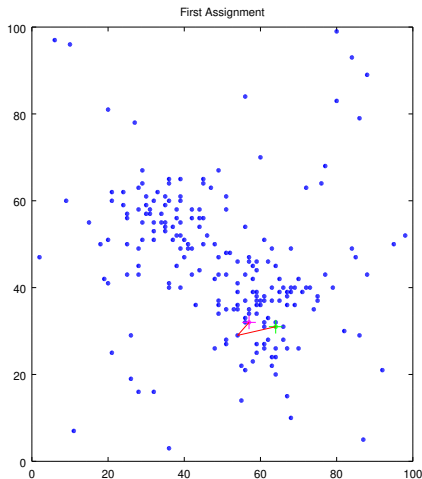
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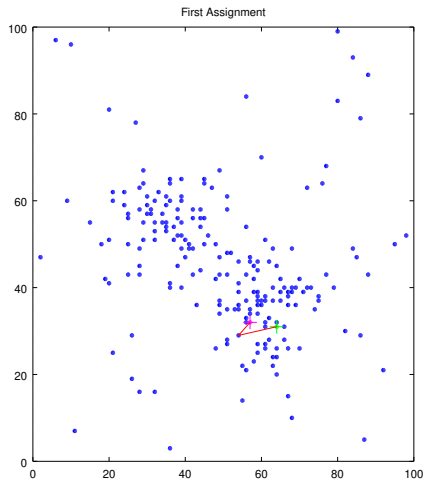
K-Means Step-by-Step

1. Randomly selected 2 points.
2. The magenta and the green points make up our two cluster groups.
3. While data remains...
 - 3.1 Randomly select a new point (blue dot with magenta and green lines connecting).
 - 3.2 Find the distance to the centroids (+).
 - 3.3 Assign to the closest.
 - 3.4 Remove that one and repeat.



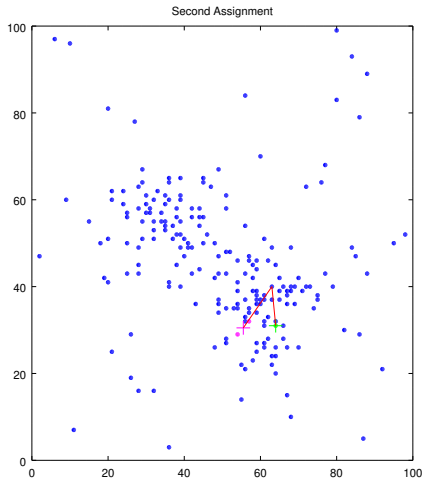
K-Means Step-by-Step

Let's pause for a second... you and I might say that these magenta and green points live in the same cluster, right? That's okay! Let's run with it...



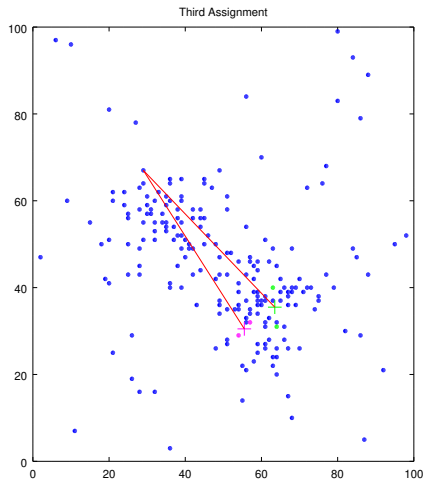
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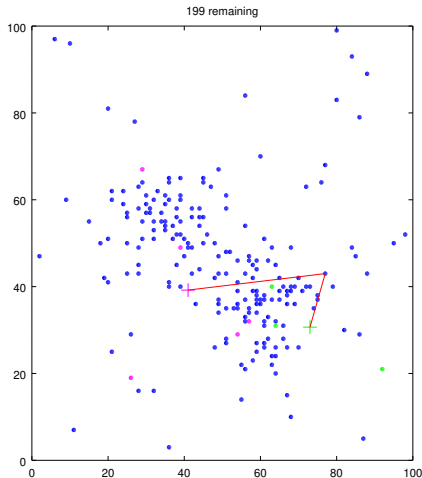
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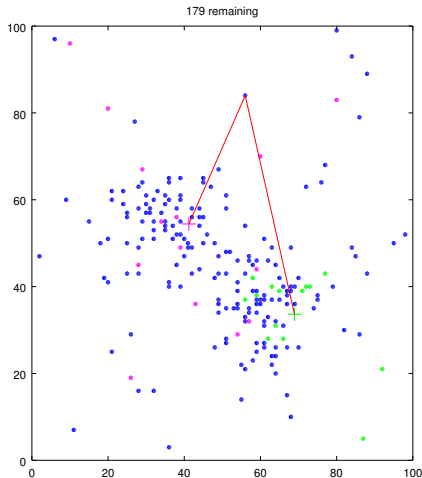
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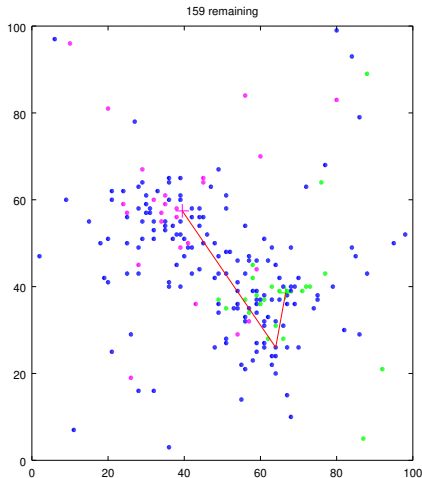
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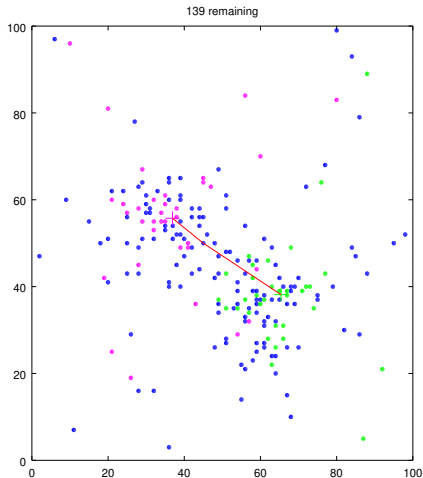
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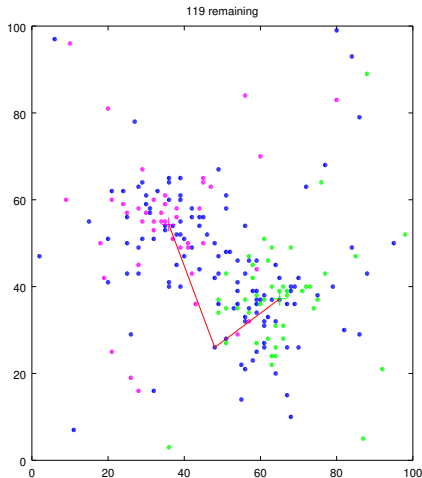
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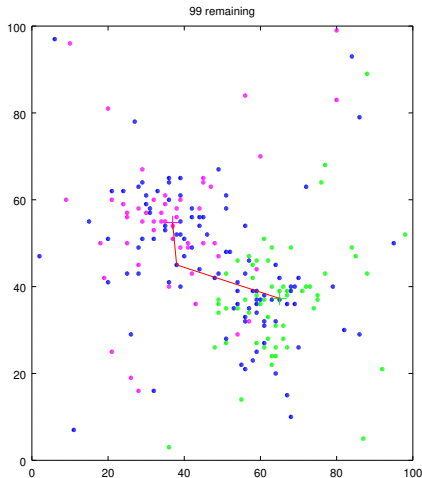
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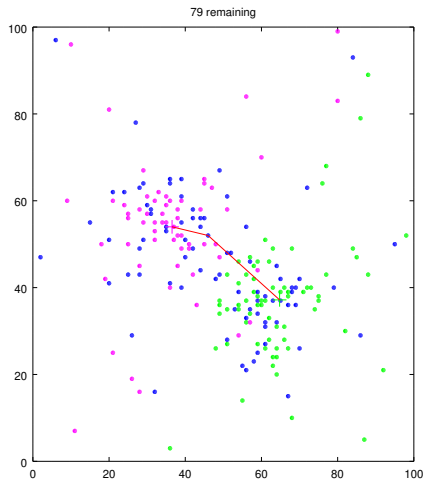
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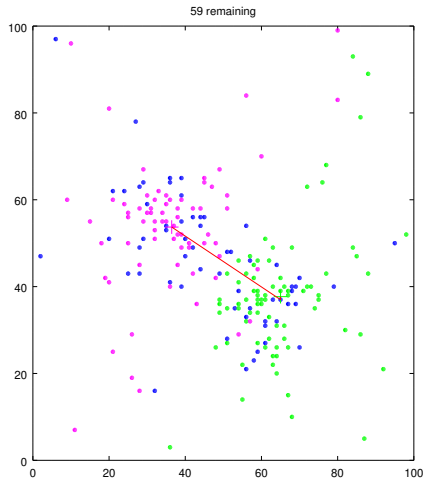
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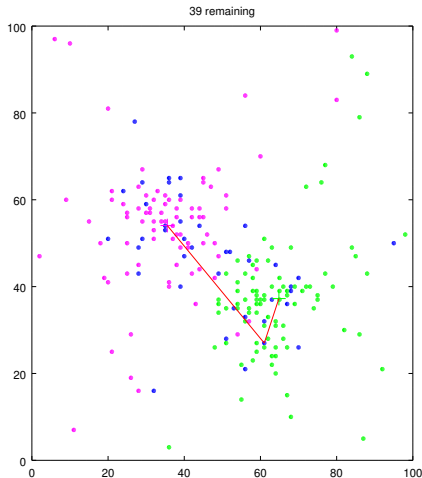
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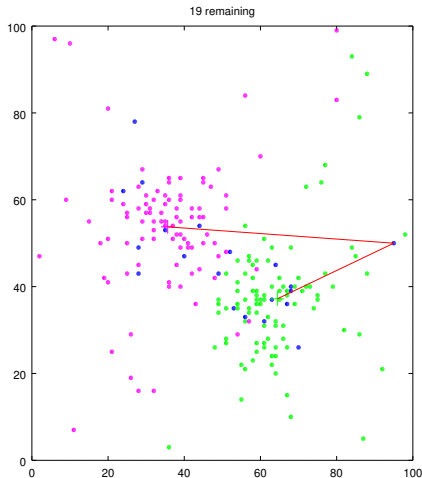
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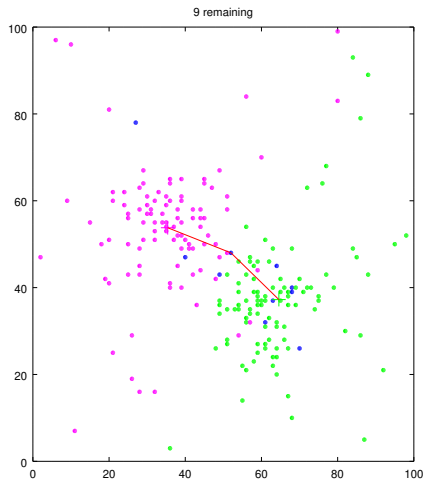
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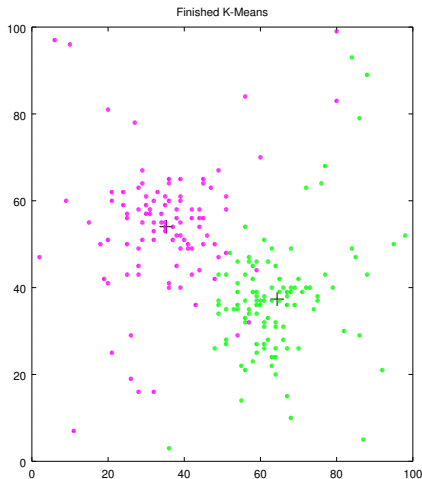
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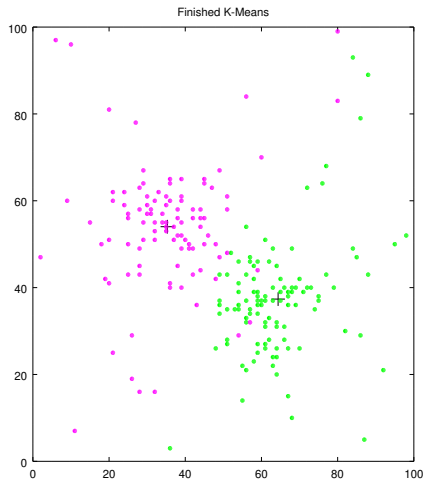
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K-Means Step-by-Step

We have pulled them apart!
Yes, some magenta dot live inside of the
bounds of the green, and vise-versa...
why?



K-Means Step-by-Step

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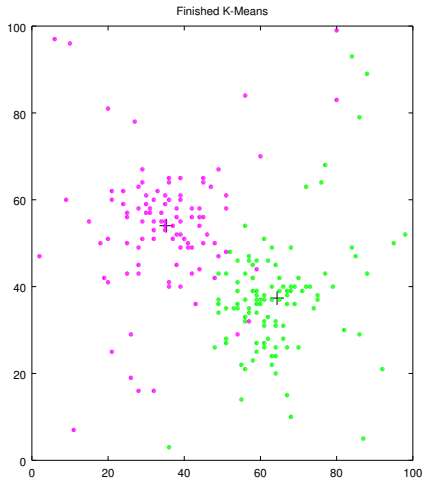
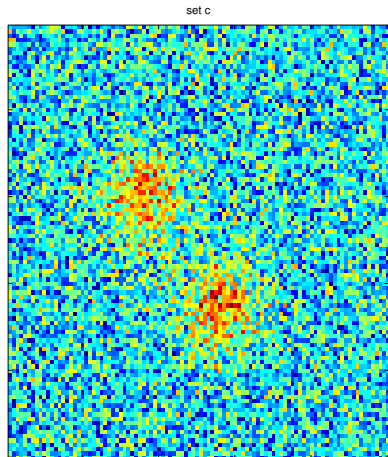
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Comparing K-Means and
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K-Means Step-by-Step

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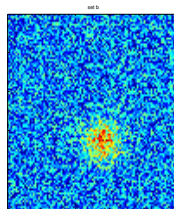
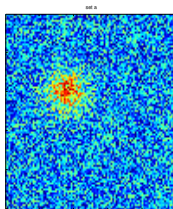
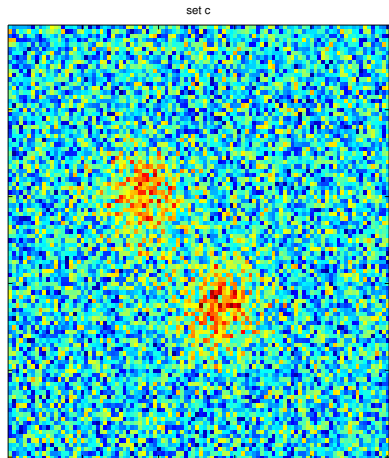
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These might be two different datasets,
that are clustered near each other.
Could there be more? (2, 100, 200?)
What might be some of the problems
with k-means? (and the fix?)
What are some advantages of k-means?

K-Means Recap

- ▶ For some new data points, consider what cluster centroid it is closest to, for assignment
- ▶ k-means is an unsupervised learning method
- ▶ nothing is known about classification of data points in the group
- ▶ we seek to identify groups that are similar

K-Nearest-Neighbors (KNN)

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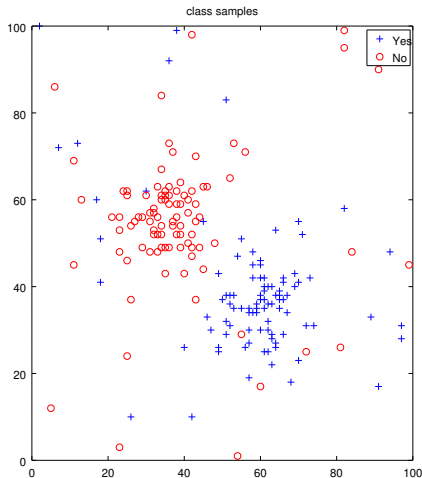
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Suppose we have labels...
This could be Y/N, or Type 1, 2, 3, etc.
We have assigned some red, and some blue.
Oh, and this is that same dataset!



K-Nearest-Neighbors (KNN)

- ▶ This time, assignment is decided on the N closest data points.
- ▶ Identify what the distribution of those data point classifications. This is a vote!
- ▶ Supervised learning, as you're using the classification of other data to determine the classification of the new data points.

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KNN How-To

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1. For some point, select the N closest other data points.
2. Find the classification with the highest *observed* probability for this set.

KNN How-To

K-Nearest Neighbors

In theory we would always like to predict qualitative responses using the Bayes classifier. But for real data, we do not know the conditional distribution of Y given X , and so computing the Bayes classifier is impossible. Therefore, the Bayes classifier serves as an unattainable gold standard against which to compare other methods. Many approaches attempt to estimate the conditional distribution of Y given X , and then classify a given observation to the class with highest *estimated* probability. One such method is the *K-nearest neighbors* (KNN) classifier. Given a positive integer K and a test observation x_0 , the KNN classifier first identifies the K points in the training data that are closest to x_0 , represented by $\mathcal{N}_0$. It then estimates the conditional probability for class j as the fraction of points in $\mathcal{N}_0$ whose response values equal j :

$$\Pr(Y = j|X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(y_i = j). \quad (2.12)$$

Finally, KNN applies Bayes rule and classifies the test observation x_0 to the class with the largest probability.

KNN How-To

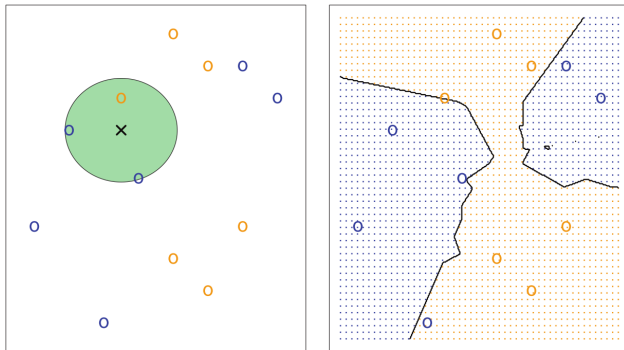


FIGURE 2.14. The KNN approach, using $K = 3$, is illustrated in a simple situation with six blue observations and six orange observations. Left: a test observation at which a predicted class label is desired is shown as a black cross. The three closest points to the test observation are identified, and it is predicted that the test observation belongs to the most commonly-occurring class, in this case blue. Right: The KNN decision boundary for this example is shown in black. The blue grid indicates the region in which a test observation will be assigned to the blue class, and the orange grid indicates the region in which it will be assigned to the orange class.

KNN How-To

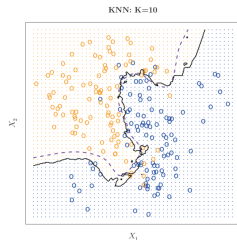


FIGURE 2.15. The black curve indicates the KNN decision boundary on the data from Figure 2.13, using $K = 10$. The Bayes decision boundary is shown as a purple dashed line. The KNN and Bayes decision boundaries are very similar.

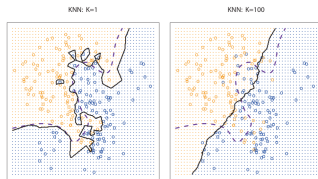


FIGURE 2.16. A comparison of the KNN decision boundaries (solid black curves) obtained using $K = 1$ and $K = 100$ on the data from Figure 2.13. With $K = 1$, the decision boundary is overly flexible, while with $K = 100$ it is not sufficiently flexible. The Bayes decision boundary is shown as a purple dashed line.

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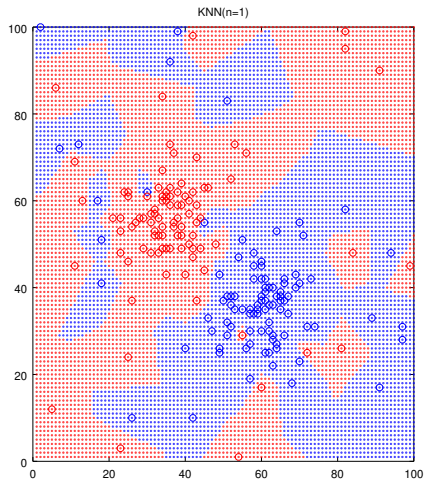
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KNN How-To

With our example data



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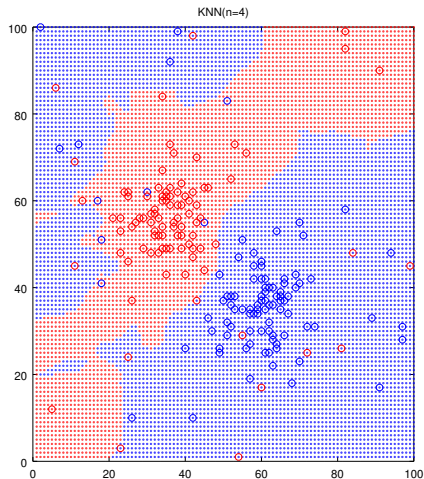
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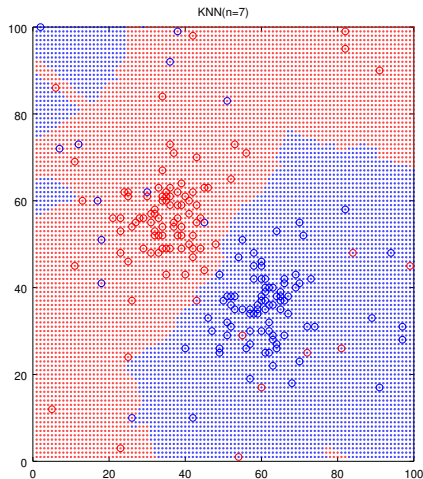
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With our example data
See how this is averaging the boundaries
out?



KNN How-To

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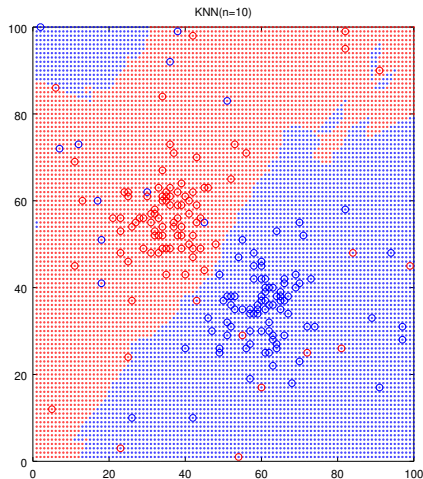
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With our example data
Almost a single dividing line. Almost.
But that's not really the goal.



Comparing K-Means and KNN

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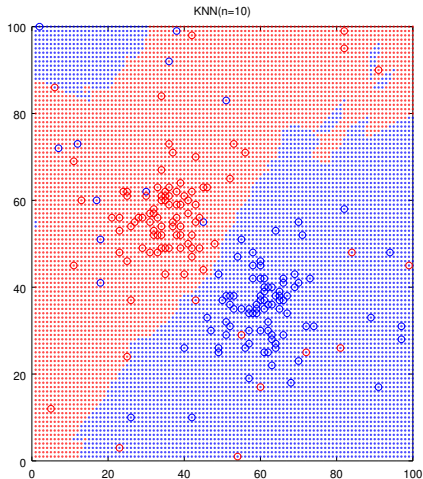
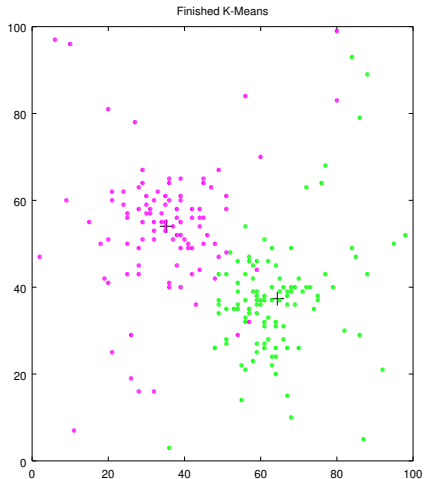
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If we introduced a new data point, then ... ?

Comparing K-Means and KNN

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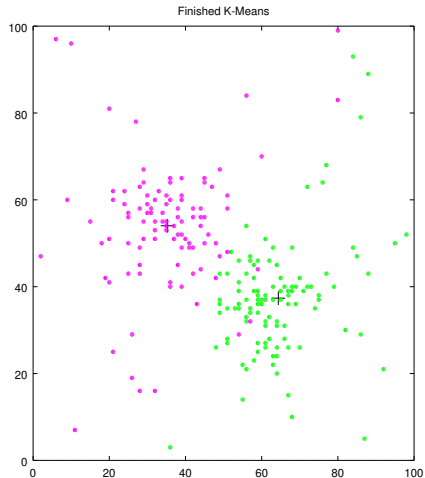
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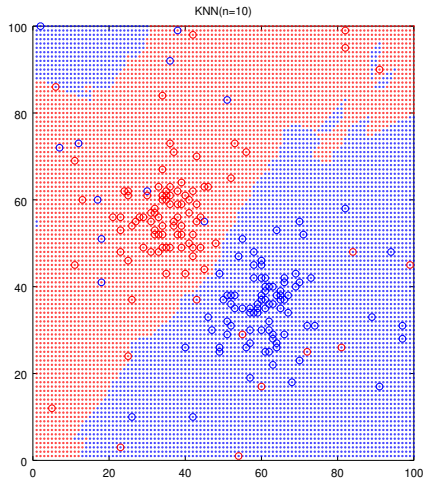
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K-Means pulls data apart, after you have the data.



Use the labeled data, can be used as a live model.